



chapter 3

Taxes and Tax Systems

Tax Base

the base of any object on which the tax due is computed, payable to a

Tax Rate

a percentage rate applied on a tax base to determine a taxpayer's tax liability

(a) Marginal tax rate

- a tax rate applied on the last dollar of taxable income
- Your marginal tax rate corresponds to the highest tax bracket your last dollar of taxable income

Tax rate

(b) Average tax rate

- a tax rate applied to each dollar of taxable income

(c) Effective tax rate

- a tax rate applied to each dollar of economic income

Example

Consider the following data

Monthly Income	Tax Rate	Incremental Amount
0-3,000	--	3,000
3,000-10,000	10%	7,000
10,000-30,000	20%	20,000
Above 30,000	30%	--

Assume a taxpayer's monthly income is \$80,000.

$$MTR=30\% \quad ATR=25.6\% \quad ETR=24.6\%$$

Tax rate

- Which tax rate is the actual percentage you pay on the entire taxable income and gives a more accurate picture of your actual tax burden?
- Ans. Effective tax rate

3 Tax Rate Structure

- the composition of tax base and tax rate(s)

Tax Base	Proportional	Progressive	Regressive	Digressive
0-1,000	5%	--	30%	--
1,000-3,000	5%	10%	25%	10%
3,000-7,000	5%	15%	20%	18%
7,000-12,000	5%	25%	15%	24%
over 12,000	5%	40%	10%	27%

- four types of tax rate structure
 - ❑ Proportional
 - ❑ Progressive
 - ❑ Regressive
 - ❑ Digressive

(1) Proportional (Flat) Tax Rate Structure

- ❑ taxes every taxpayer with the same flat tax rate

(2) Progressive (Graduated) Tax Rate Structure

- ❑ taxes the higher income groups at higher tax rates & lower income groups at lower tax rates

(3) Regressive Tax Rate Structure

- ❑ taxes the higher income groups at lower tax rates & lower income groups at higher tax rates

(4) Digressive (Mild) Tax Rate Structure

- ❑ similar to progressive tax rate structure but tax rates increase at a decreasing rate

4 Burden of a Tax

1) **Impact**

who is originally responsible by tax law to pay tax to the government

2) **Incidence**

the ultimate resting place of a tax burden

3) **Shifting**

the process by which the money burden of a tax is transferred from one person on whom the tax is initially imposed to another person who ultimately bears the tax burden

5 Direct Taxes Vs. Indirect Taxes

Direct Taxes	Indirect Taxes
➤ whose impact and incidence fall on the same person	➤ whose impact and incidence may fall on different persons
➤ cannot be shifted to others in any form	➤ can be shifted to others in the form of price
➤ levied on incomes and properties of persons	➤ levied on consumptions or expenditures of persons

5 Direct Taxes Vs. Indirect Taxes

Comparison Item	Direct Taxes	Indirect Taxes
<i>Base of levy</i>	are levied on incomes and properties	are levied on goods and services
<i>Equitability</i>	progressive in nature b/c they are based on ability-to-pay principle	regressive in effect b/c the rich and the poor will pay equal tax on goods
<i>Administrative efficiency</i>	costs of collecting direct taxes to the gov't are relatively low	costs of collecting indirect taxes to the gov't are relatively high
<i>Certainty</i>	are certain b/c taxpayers know how much, when and on what basis they pay	are uncertain b/c of rise and fall of prices of goods and services
<i>Distortion of economy</i>	have the effect of reducing demand and prices	cause for inflationary forces through high prices
<i>Popularity</i>	are generally opposed by taxpayers as they are directly imposed on the taxpayers	are paid in terms of price of goods & taxpayers do not feel the extra tax burden

6 Ad-valorem Taxes Vs. Unit Taxes

(i) Ad-valorem Taxes

- are taxes which are levied on goods and services according to their values

(ii) Unit Taxes/Specific Taxes

- are taxes which are levied on goods and services at a fixed amount based on measurements other than the value of the goods and services

6 Ad-valorem Taxes Vs. Unit Taxes ...

Comparison Item	Ad-valorem Taxes	Unit Taxes
1) Base of levy	value of items to be taxed	any measurement other than the value of items to be taxed
2) Convenience	difficult to administer and collect	easy to administer and collect
3) Evasion	high chance of tax evasion	low chance of tax evasion
4) Revenue yield	high during inflation but low during deflation time	static during and inflation deflation time
5) Certainty	difficult to predict revenue yield	easy to predict revenue yield
6) Equitability	equitable	less equitable

7 Types of Taxes

7.1 Taxes on Income

7.2 Taxes on Expenditure

7.3 Taxes on Properties

7.1 Taxes on Income

Income Taxes?

- taxes levied and charged by governments on income of persons
 - ✓ tax on employment income, tax on business income, tax on rental income, tax on dividend income, tax on interest income, tax on royalty income, tax on capital gain ...

Global Income Tax Vs. Scheduler Income Tax

(i) Global Income Tax

- levied on the global income of a taxpayer

(ii) Scheduler Income Tax

- levied on the scheduler income of a taxpayer

Individual Income Tax Vs. Corporate Income Tax

(i) Individual Income Tax/Personal Income Tax

- is an income tax levied on the income of an individual for a given tax period

(ii) Corporate Income Tax/Company Income Tax

- is an income tax levied on the profit of a corporation/company for a given tax period

7.2 Taxes on Expenditure

- taxes levied and charged by governments on goods and services
 - value added tax
 - turnover tax
 - sales tax
 - excise tax
 - customs duty
 - sin tax
 - etc.

7.2.1 Value Added Tax (VAT)

What is VAT?

- is a consumption/expenditure tax levied on the value added to goods and services by importers, manufacturers and traders at each stage of the production and distribution process

Features of VAT

- ✓ indirect tax
- ✓ consumption tax
- ✓ comprehensive tax/broad based tax/multi-stage tax
- ✓ apportioned tax/ fractionally collected tax
- ✓ neutral tax
- ✓ works on the principle of tax credit system

VAT Rates

Standard vat rate

- is a positive vat rate ($>0\%$) set by a country's tax law to charge vat on goods and services
 - Ethiopia, 15%; Kenya, 16%; Cote-Devoir, 20%; Germany, 19%; UK, 21%; Argentina, 27%; Chili, 19%; Canada, 7%

Zero vat rate

- is a vat rate of zero (nil) set by a country's tax law to charge vat on goods and services

VAT Rate Systems

(A) Single vat rate system

- a uniform vat rate on all types of goods and services
 - Ethiopia, 15%
 - easy to administer the tax
 - efficient in tax administration cost

VAT Rate Systems ...

(B) Multiple vat rates system

- different vat rates on different types of goods and services
 - Kenya
 - Standard vat rate is 16%; vat on hotel services is 7%
 - Bulgaria
 - Standard vat rate is 19%; vat on hotel services is 12%
 - Norway
 - VAT rates split into three levels: 25%, 14% and 8%
- difficult to administer the tax
- high in tax administration cost

Components of VAT

(A) Input VAT

- vat paid on purchases of goods and services

(B) Output VAT

- vat collected on sales of goods and services

(C) Net VAT

- the difference between output vat and input vat
 - Net vat liability/due if $\text{Output VAT} > \text{Input VAT}$
 - Net vat credit/refund if $\text{Output VAT} < \text{Input VAT}$

Exempt Supply Vs. Taxable Supply

Supply:

➤ goods and services available for sale; sale of goods or provision of services

(A) Exempt supply

- goods and services on which vat is not charged
 - supply of certain basic food items like milk, bread, enjera, teff, wheat, etc.
 - supply of printed books
 - provision of educational services
 - provision of medical services
 - provision of water, electricity and kerosene
 - provision of inland public transport services

Exempt supply Vs. Taxable supply ...

(B) Taxable supply

- goods and services on which vat is charged at either a standard rate or zero rate
 - standard rated supply
 - zero rated supply

(i) Standard rated supply

- goods and services on which VAT is charged at the standard rate set by the country/state, say, 15⁰% in Ethiopia
 - all taxable supplies other than zero rated supplies

Exempt supply Vs. Taxable supply...

(ii) Zero rated supply

- goods and services on which VAT is charged at 0%
 - **export of goods and services**
 - **provision of international transport services**
 - **supply of gold to NBE**

Exempt supply Vs. Taxable supply...

Illustration

Arsema and Rommel own Binda Smelting PLC. The following information belongs to Binda Smelting PLC (smelter) that purchased iron ore (raw materials) from Red-Night Mining PLC (minor). Both are vat-registered suppliers.

- VAT exclusive purchase price of iron ore (raw materials) by Binda PLCBr.100,000
- VAT exclusive selling price of steel (finished product) by Binda PLCBr.150,000
- If the steel are,
 - standard rated goods at 15%
 - zero rated goods
 - exempt goods

Exempt supply Vs. Taxable supply...

Required:- Determine the following:-

- a) The input vat (VAT to be refunded) for Binda Smelting PLC (assume the iron ores [raw materials] are standard rated supply at 15%)
- b) The cost of purchase of raw materials to Binda Smelting PLC
- c) The output vat (VAT liability) for Binda Smelting PLC
- d) The net vat payable to the tax authority or refundable by the tax authority to Binda Smelting PLC

Exempt supply Vs. Taxable supply...

i) If the steel are standard rated

a) The input vat (vat credit)

- Br.15,000 (15% of 100,000)

b) The cost of purchase of raw materials

- Br.100,000

c) The output vat (vat liability)

- Br.22,500 (15% of 150,000)

d) The net vat payable to (refundable by) the tax authority to Binda PLC

- Br.7,500 (22,500-15,000)

Exempt supply Vs. Taxable supply...

i) If the steel are zero rated

a) The input vat (vat credit)

Br.15,000 (15% of 100,000)

a) The cost of purchase of raw materials

Br.100,000

b) The output vat (vat liability)

Br.0 (0% of 150,000)

a) The net vat payable to (refundable by) the tax authority to Binda PLC

-Br.15,000 (0-15,000)

Exempt supply Vs. Taxable supply...

i) If the steel are exempt goods

a) The input vat (vat reclaimed)

- Br.15,000 but not refundable

b) The cost of purchase of raw materials

- Br.115,000 (100,000+15,000)

c) The output vat (vat liability)

- Br.-

d) The net vat payable to (refundable by) the tax authority to Binda PLC

- Br.-

Uniform Supply Vs. Mixed Supply Vs. Composite Supply

- ✓ **Uniform supply:-** occurs if a good, a service, identical goods or identical services is/are supplied/invoiced in a single VAT inclusive price
- ✓ **Mixed supply:-** occurs if a mixture of goods and/or services is invoiced together at a single VAT inclusive price; it is possible to split the mixed supply into its component parts for vat purpose
 - ✓ *For instance, supply of a television, a video player and a refrigerator in one invoice (all taxable). A supply of a stationary material (taxable) and a book (exempt).*
- ✓ **Composite supply:-** occurs if a mixture of goods and/or services is supplied together in such a way that it is not possible to split the supply into its component parts
 - ✓ *For instance, supply of an electronic item including the services of transportation and labor*

VAT Exclusive Price Vs. VAT Inclusive Price

VAT Exclusive Price

- ✓ a method that the sale transaction does not include the amount of vat
- ✓ It is a purchase or sale price before vat

VAT Inclusive Price

- ✓ a method that the sale transaction includes the amount of vat
- ✓ It is sales price including vat

Invoice-based Method Vs. Accounts-based Method

Invoice-based Method

- each vat-registered supplier/seller charges vat on his sales and passes the buyer a special invoice that indicates the amount of tax charged

Accounts-based Method

- the tax is calculated on the value added, which is measured as the difference between revenues and allowable purchases

Accrual-basis Method Vs. Cash-basis Method

Cash-basis Method

- when payment is received for the sale of goods or services, the revenue including vat is recorded as of the date of the receipt of cash from the customer, no matter when the sale will be made
- purchase and expense are recorded as of the payment date, regardless of when the expense incurred

Accrual-basis Method

- matches revenues to the time period in which they are earned
- matches expenses to the time period in which they are incurred

VAT Return and VAT Accounting Period

VAT Return

- a declaration form to be filled by a supplier and submitted to the tax authority at the end of each vat accounting period

VAT Accounting Period

- the time period a supplier is required to furnish a vat return to the tax authority
- the period of vat payment to the tax authority

7.2.2 Excise Tax

What is Excise Tax?

- an excise tax (sometimes called a duty of excise or a special tax) may be defined as a tax levied on a certain good produced for sale within the country
- Excise tax is distinguished from sales tax or value added tax in three ways:
 - excise tax typically applies to a narrower range of products;
 - excise tax is typically heavier, normally accounts for higher fractions (sometimes half or more) of the tax base of the targeted products; and
 - excise tax is typically specific (i.e., a fixed amount of tax per unit of measure; e.g. Br.10 per gallon), whereas a sales tax or value added tax is ad-valorem, i.e. proportional to value (a percentage of the price in the case of a sales tax, or of value added in the case of a value added tax)

Excise Tax in Ethiopia?

- ‘**Excise tax**’ is ‘an indirect tax levied on certain types of locally manufactured or imported goods that are hazardous to health and cause to social problem, basic goods which are demand inelastic, and luxury goods.’

Why?

- to improve the government revenue by imposing excise tax on luxury goods as well as basic goods which are demand inelastic; and
- to reduce and regulate the consumption of goods that are hazardous to peoples’ health and which are cause to social problems by imposing excise tax on such goods, thereby increasing their prices

7.3 Taxes on Properties

What?

- are taxes levied and charged by governments on properties belonging to persons

Types

- wealth tax, inheritance tax, gift tax, estate tax, land property tax, etc.

7.3.1 Wealth Tax

What ?

- a tax levied and charged on the value of net wealth/net assets belonging to a person (taxpayer) on the valuation date based on specific or range of rates
 - net wealth/net assets
 - value
 - person
 - valuation date
 - tax rate

Why ?

- wealth inequality

8.3.1 Wealth Tax ...

- Several countries, including Austria, Denmark, Germany, Netherlands, Finland, Sweden, and Greece, have abolished their wealth tax in recent years
- Others countries, including Britain and Belgium, have never had such a tax
- But Britain proposed new wealth tax reform by 2021

Countries currently with wealth tax:-

- **India:** a rate of 1% of net wealth
- **France:** a progressive rate from 0 to 1.8% of net assets
- **Spain:** a progressive rate from 0.2 to 2.5% of net assets
- **Iceland:** a rate of 1.5% on net assets
- **Norway:** a rate of 0.7% (municipal) and 0.3% (national) on net assets
- **Switzerland:** a progressive wealth tax with a maximum of 1.5% on net assets

7.3.2 Inheritance Tax

- is a tax payable by a beneficiary when the beneficiary inherits money or property from the estate of a deceased person

Example

The Ethiopian government charges a 5 percent tax on all inheritances larger than \$2 million. Therefore, if your father or friend leaves you \$5 million in his will, you will pay inheritance tax on the \$3 million, which is \$150,000 (5% of 3,000,000).

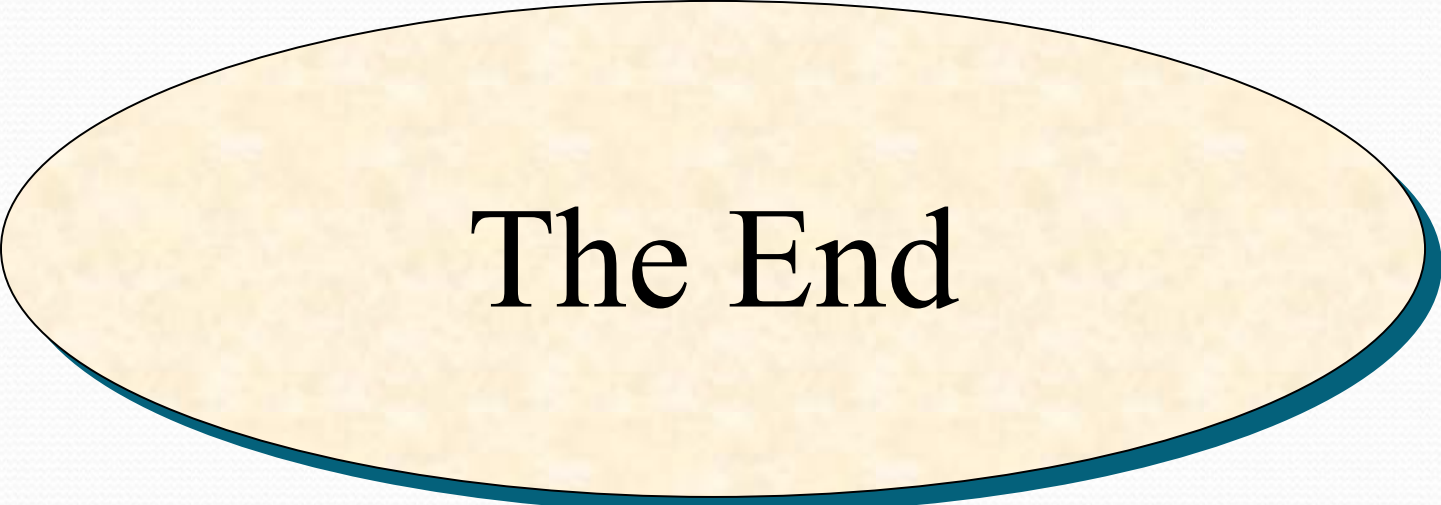
7.3.3 Gift Tax

What?

- is a tax on the transfer of property (or anything of value) by one individual to another while receiving nothing, or less than full value, in return.

Who?

- giver or receiver of the gift?



The End